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HHS Public Access

Author manuscript

Circ Heart Fail. Author manuscript; available in PMC 2026 March 17.

Published in final edited form as:

Circ Heart Fail. 2026 April ; 19(4): e014013. doi:10.1161/CIRCHEARTFAILURE.125.014013.

MUTTON-HF: Rationale and Design of a Study of an Indigenous Food is Medicine Intervention

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Abstract

Background: Nutrition insecurity is a major driver of poor cardiovascular health in Indigenous communities. Medically tailored meals (MTMs) may improve heart failure outcomes and quality of life. There is growing momentum among Indigenous communities to reclaim traditional pre-contact foods to improve cardiovascular health. In this context, utilizing community-based-

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Disclosures: N/A

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participatory methods, we designed MUTTON-HF (Medically Utilized Tailored Traditional foods to Optimize Nutrition in Heart Failure)— an Indigenous culturally and medically tailored (CMTM) program that locally sources Native produce and meat and incorporates traditional Diné (Navajo) foods and recipes.

Methods: The MUTTON-HF trial is a 2-center, pragmatic, open-label randomized controlled trial to determine the efficacy and safety of MUTTON-HF among Diné patients with heart failure and hospitalization, or emergency room (ER) visit in the last 12 months. Two hundred and four subjects at 2 Indian Health Service sites will be randomized in a 1:1: stratified fashion by gender, age (<65 vs ≥ 65 years) and left ventricular ejection fraction (<50% vs ≥50%). Study subjects will receive either CMTMs that incorporate traditional Diné recipes and foods or usual dietary advice for 8 weeks. Data analysts will be blinded to group assignment until study completion.

Results: The primary efficacy end point is the proportion of patients with a hospitalization or ER visit (all-cause) within 90 days. Secondary outcomes will include hospitalizations or ER visits for heart failure specifically; Kansas City Cardiomyopathy Questionnaire scores for health-related quality of life; measures of food insecurity; Indigenous nourishment; diet quality; financial strain; and clinical biomarkers from study enrollment to 8 weeks.

Conclusions: Community-based interventions that leverage the protective assets of Indigenous communities are needed to improve cardiovascular health among Native populations. This pragmatic randomized controlled trial will target a rural tribal population and be the first to evaluate the efficacy of an Indigenous food is medicine intervention to improve Indigenous cardiovascular health outcomes for patients with heart failure.

Registration: Clinicaltrials.gov NCT06549699 (www.clinicaltrials.gov/study/NCT06549699)

Introduction

The ongoing impacts of settler colonialism produce adverse structural drivers – particularly, nutrition insecurity – that contribute to cardiovascular health disparities among Indigenous populations. [1–4] Traditionally, Indigenous people lived a healthy lifestyle with high rates of physical activity and consumption of traditional healthy foods. However, Indigenous genocide, governmental relocation policies, broken treaty obligations, decades of harmful environmental policy and environmental injustice, and targeted and sustained community disinvestment combine to fuel poverty and nutrition insecurity among Native communities. [1–4] On the Navajo Nation, nutrition insecurity is endemic and contributes to poor cardiovascular health [5]; a recent qualitative study of Diné (Navajo) patients living with HF determined that nutrition insecurity was the major barrier to achieving optimal cardiovascular health.[6]

Dietary factors play a key role in both clinical and patient-reported HF outcomes. There is increasing evidence that Food is Medicine Programs, such as medically tailored meals, may improve HF outcomes and disease-specific quality of life.[7–9] Additionally, there is growing momentum among Indigenous communities to advance food sovereignty and reclaim traditional pre-contact foods to improve health.[10–21] Leveraging the innumerable assets of Indigenous communities is key to advancing Indigenous health.[1] ‘Food is medicine’ programs may address Indigenous determinants of health by bolstering the

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inherent strengths and resources of Indigenous communities through reclaiming traditional foods, strengthening cultural connections to the healing properties of food; encouraging intergenerational transfer of knowledge including how to cook with traditional ingredients, and respecting the deep connection to ancestral and other lands by engaging local and traditional farmers and ranchers, all of which are central to holistic health and well-being.[1] [22]

Given this, utilizing community-based participatory methods, we designed and piloted Mutton-HF- a locally and Native sourced medically and culturally tailored meal (CMTM) delivery program incorporating traditional Diné foods and recipes.[23] We recently completed a pilot feasibility trial of the MUTTON-HF intervention, which demonstrated that the intervention was feasible, despite patients separated by vast distances on the Navajo reservation (>100 miles) and highly acceptable.[24] Additionally, rates of food insecurity, cultural connectedness, quality of life and symptom measures were significantly improved after receiving one month of the intervention, and there was significant weight loss among those with baseline obesity.[24] For this reason, we are undertaking a trial of Medically Utilized Tailored Traditional food to Optimize Nutrition in Heart Failure (MUTTON-HF) to test the hypothesis that a locally sourced Indigenous CMTM program will lead to lower incidence of hospitalizations and ER visits when compared with usual care.

Methods

The Mutton-HF Clinical Trial

The full study protocol is available at wwww.muttonhf.com and clinicalstrials.gov (NCT06549699) [23][25]; Figure 1 shows the study flow and design. MUTTON-HF is a pragmatic, randomized, open-label randomized controlled trial at two Indian Health service (IHS) sites that tests the effectiveness of a Native and locally sourced medically tailored meal program for HF. This study occurred as part of an IHS Office of Quality Innovations Award to improve the quality of cardiovascular care at two IHS sites in Navajo Nation and was funded by the American Heart Association Health Care by Food Initiative. This study was approved by the Navajo Nation Human Research Review Board (NNR 24.525) and Mass General Brigham. This trial was conducted in accordance with the principles of the Declaration of Helsinki and will follow the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines. Patient consent was obtained either in-person or verbally over the phone. The study began on 05/15/2025 and completed enrollment on 08/02/25.

Study Setting

This study is being performed at 2 IHS ambulatory clinics in Navajo Nation, one of which is the largest IHS facility serving the rural Navajo Nation, the largest Indigenous tribe in the US.[26,27] Located 45 miles North of the first site, the second site is a smaller satellite IHS clinic which was established to serve patients living remotely on the Navajo reservation. These two facilities serve a catchment area of >75-mile radius (Figure 2).[26] Of the 27 000 square miles of land on the Navajo Nation, the largest US reservation, there are only 13 grocery stores and approximately 110 smaller stores which are limited in fresh produce and healthy food options.[28]

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Inclusion and Exclusion Criteria

In this pragmatic trial, we included patients with a HF diagnosis (I50.*) who were receiving care at one of the two IHS facilities (defined as at least one clinical encounter and one prescription in the last 12 months) and had either an emergency room visit or hospitalization in the last 12 months. Patients on hospice, living at skilled nursing or acute rehabilitation facility or unable to provide consent were excluded.

Intervention

Design of the Mutton-HF intervention is described in detail elsewhere.[24] Briefly, we used community-based participatory methods to design an Indigenous Food is Medicine intervention in which meals were created by a local Diné dietician and culinary expert to incorporate traditional Diné foods and recipes (Figure 3).[29] Meals were medically tailored to meet American Heart Association's sodium-restricted Dietary Approaches to Stop Hypertension (DASH) recommendations.[30][31] We partnered with local Diné farmers and ranchers to locally source meat and produce not only to strengthen local food systems and advance food sovereignty, but also to increase acceptability and accessibility of the meals and the likelihood of long-term adoption of recipes after the intervention period. To align with cultural practices and beliefs, meat was obtained with the goal of whole animal utilization.[21] We partnered with Tocabe, a Native-run kitchen to produce the medically tailored frozen meals.[32] Given preliminary data demonstrated >95% of patients do not have physical mailboxes (and therefore the ability for mail deliveries at home) on the reservation, a delivery distribution network was established. Meals were delivered to a central food hub for patient pick-up. The satellite clinic served as a second hub for pick-up. For individuals who were unable to pick up at either location, we tailored distribution plans, distributing meals to "mini hubs" (e.g. Chapter houses or community centers) on the reservation for pick up, or providing direct-to-patient delivery in partnership with Community Health Representatives (Figure 3). Patients were provided with the necessary household infrastructure to successfully participate in the intervention including microwaves, freezers, etc. Patients without stable electricity were provided a propane-fueled refrigerator and stove, as well as a dedicated filled propane tank to power appliances.

Patients were randomized to either the intervention or control arm. Both arms received usual care. Patients in the intervention arm received 14 medically and culturally tailored traditional Diné meals weekly (2 meals daily) for 60 days. Upon study enrollment, both groups received an educational booklet with recommendations for a heart-healthy and DASH diet. At the end of the outcome assessment period, both groups also receive a cooking class to learn how to sustain healthy behaviors and learn to cook favorite meals from the program. Given community preferences and ethical concerns, all patients in the control arm can opt into receiving 8 weeks of the intervention meals after the outcome assessment period and patients can be enrolled in the study and opt out of survey and laboratory data collection at any point during the study.

Enrollment Period

To allow time to coordinate the operational and logistical infrastructure for meal delivery based on patient location/community and delivery mechanism, and procure necessary home

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appliances, patients were consented and enrolled up to 45 days prior to study start (t=0). During this time, baseline assessment with surveys and lab work was conducted. All patients started the study period (and outcome assessment period) at the same t=0, i.e. the date of the first meal pick-up for the intervention group. Any events that occurred in the −45 to −1 days prior to the intervention period will not be included as an outcome.

Outcomes

All primary and secondary outcomes are summarized and defined in Table 1. The primary outcome is the proportion of patients who have a hospitalization or ER visit for any cause within 90 days.

Secondary Outcomes

Secondary outcomes (Table 1) include the proportion of patients with HF hospitalizations specifically (primary diagnosis), the proportion of patients with ER visits for volume overload or HF specifically, change in blood pressure, weight and body mass index at 60 days from baseline, 12-item Kansas City Cardiomyopathy Questionnaire (KCCQ) Summary Score (and its components including physical limitation score, symptom frequency score, quality of life score, and social limitation score)[33], and change in the following lab parameters and biomarkers from baseline to 60 days: albumin, prealbumin, creatinine, NT-proBNP, HbA1c, Total cholesterol, LDL-C, HDL-C, Triglycerides, CRP, urine albumin: creatinine ratio. All patients were provided with a home blood pressure cuff (Omron 5 series) and scale for blood pressure and weight measurements.

Additional secondary outcome measure include diet quality, assessed using the 10-item Diet Summary Questionnaire (DSQ) [34] with an addition question to assess traditional Diné food intake (*During the past month, how often did you eat traditional Diné foods?)* at 60 days. We will also evaluate changes in rates of food insecurity at 60 days utilizing the USDA Adult Food Security: Six Item short Form survey (USDA FSSM 6-Item), [35] Indigenous cultural connectedness utilizing a 6-question modified version of the validated Cultural Connectedness Scale-California (CCS-CA), the Indigenous Nourishment scale (Measure A) at 60 days[36][37], and changes in financial stress at 60 days using the Financial Stress Scale.[38] The CCS-CA is a validated score, developed by Indigenous scholars to measure cultural connectedness in Indigenous populations, and has been shown to correlate with both physical and mental health and be a social determinant of health.[39] This scale has previously been modified by our group for use in Diné populations. The Indigenous Nourishment Scale was developed through community-based methods in Indigenous communities to encompass multiple dimensions of nourishment (physical, spiritual, emotional and relational).[37] While the CCS-CA measures overall cultural connectedness, the Indigenous Nourishment Scale focuses on strength-based measures of nutrition to add additional insight into nourishment and the relational connection to the land and community through food specifically. These two scales offer an Indigenous framework to allow for strength-based health measurements rooted in Indigenous perspectives.

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Intervention characteristics:

To evaluate the impact on local and Indigenous food systems, we will evaluate the proportion of meals that include locally sourced ingredients, proportion sourced by Native suppliers/farmers, and those by Diné producers/suppliers/farmers specifically. We will evaluate the percentage of meals successfully distributed to the patient. Specific dietary consumption is assessed by 3-day food diaries during weeks 1 and 8. Patient satisfaction will be evaluated post-intervention by patient rating of the program (rating range, 1–10; response to the question, “How likely is it from 1–10 that you would recommend this program to a community member?”), which will also be used to calculate the Net Promoter Score. We will evaluate fidelity and compliance with a survey at the end of the program to assess how many of the meals were consumed by the patient. We will also assess safety of the intervention with adverse events as described in Table 1.

Exploratory Outcomes

Additional exploratory outcomes will include time to event for the primary outcome, and prescription adherence rates for all medication as well as guideline-directed medical therapy, specifically. In addition to 60-day measurements, we will also collect weight measurements at 30 days and 90 days. We will evaluate how much physical activity/exercise patients are participating in weekly at baseline, and at 60 days as well as patient-reported health status.

Implementation Outcomes

We will also perform semi-structured interviews of non-patient stakeholders (i.e. farmers that supplied produce, livestock farmers, food pantry) to assess acceptability and feasibility of providing produce for the meals, as well as the degree to which the program contributed to strengthening the local food system and advancing food sovereignty. We will also perform semi-structured interviews of primary care providers and a subset of patients to assess acceptability, as well as explore patient themes related to cultural connectedness and determinants of Indigenous health, as well as explore behavioral change trajectories and adoption of recipes from the program. Semi-structured interviews will be guided by the Consolidated Framework for Implementation Research (CFIR), and we will explore multiple constructs within each CFIR domain that are hypothesized by the study team and based on existing literature to be relevant to Indigenous holistic health and nourishment, as well as acceptability and feasibility of the program.[40]

Randomization and Blinding

Participants were randomized in a 1:1 ratio to intervention or usual care using permuted block randomization (block size = 6), stratified by left ventricular ejection fraction (<50% vs. ≥50%), age (<65 vs. ≥65 years), and sex. For participants from the same household, the first enrollee was randomized, and subsequent members were assigned to the same arm to reduce contamination.

Neither participants nor providers were blinded to allocation. Outcomes will be ascertained from patient surveys and electronic health records (EHR). Data analysts and study staff extracting outcomes from the EHR and recording outcomes will remain blinded to treatment assignment.

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Statistical Analysis Plan

Intention-to-Treat Analysis: The primary follows the intention-to-treat (ITT) principle, corresponding to a treatment-policy strategy for intercurrent events, consistent with ICH E9(R1).[41] Under this estimand, all randomized participants will be analyzed according to their assigned treatment group, regardless of adherence to the intervention, protocol deviations, or early discontinuation. Expected intercurrent events include non-adherence to assigned meals, partial adherence, receipt of non-study meals, and loss to follow-up prior to 90 days. For the primary ITT estimand, these events will not alter treatment assignment or inclusion in the analysis; participants experiencing such events will remain in the analysis denominator, with outcomes ascertained from the EHR when available. Between-group differences in the primary outcome and dichotomous secondary outcomes will be assessed using $\chi 2$ tests and continuous secondary outcomes with 2-sample t tests. Within-group randomization to week 8 changes in continuous outcomes including laboratory evaluation and biomarkers, Indigenous nourishment scale, financial stress scale, and KCCQ scores, will be evaluated using paired t-test.

Per-Protocol Effect (Adherence) and Compliance Adjustment

Estimand. To complement the ITT estimand, we will additionally conduct pre-specified per-protocol and compliance-adjusted analyses, which target a distinct estimand reflecting the effect of sustained adherence to the assigned strategy through days (medically tailored meals vs. usual care). Unlike a completers-only analysis, which is subject to strong selection bias, this approach uses causal methods to approximate the effect had all participants fully adhered to their randomized assignment [42]. In these analyses, participants will be administratively censored at the time of protocol deviation or loss to follow-up, and causal methods (marginal structural models with inverse-probability weighting) will be used to address time-varying adherence and informative censoring.

Adherence definition and grace period. Adherence will be measured from delivery logs and participant meal-use reports, aggregated to weekly intervals. “Adherent” is defined as the ≥70% adherence to meal delivery and consumption and completion of baseline and post-intervention assessments and lab parameters.

Implementation. We will emulate a per-protocol comparison by administratively censoring participants at the time of protocol deviation (e.g., falling below the adherence rule). Analyses will use marginal structural models with stabilized inverse-probability weights (IPW) for adherence and informative loss to follow-up.[42] Models will adjust for baseline covariates (age, sex, LVEF category, comorbidities, baseline diet quality) and time-updated predictors measured prior to deviation (e.g., interim hospitalization, weight change, medication use). The primary contrasts will be standardized risks (risk difference and ratio) at 60 days, estimated from weighted pooled logistic regression for binary outcomes and weighted generalized linear models for continuous outcomes. Robust (sandwich) variance will be used, clustering by household (and site, if indicated).

As-treated and principal-stratum complements. As a complement, we will conduct an exploratory as-treated analysis, modeling actual meal consumption as a time-varying

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exposure with IPTW for time-varying confounding. We will also estimate a complier average causal effect (CACE) using randomization as an instrumental variable, under standard IV assumptions [43], to quantify the causal effect among latent compliers.

Missing adherence data. Missing adherence and time-updated covariates will be addressed using multiple imputation prior to weight construction.[44] Sensitivity analyses will repeat weighting using inverse probability of missingness instead of imputation.

Sensitivity analyses. To assess robustness, we will (i) vary the adherence threshold (e.g., 60% and 80%), (ii) alter specifications and truncation levels for the IP weights, (iii) treat control-arm “contamination” (receipt of meals outside the trial) as either nonadherence or allowable deviation, and (iv) repeat per-protocol analyses without imputation (complete cases only).

Given the low risk of the intervention, prior pilot data supporting safety of the intervention [24], and short duration of the intervention, no interim safety analyses are planned.

Qualitative analyses: Semi structured interviews will be audio-recorded and transcribed for thematic analysis. Transcripts will be analyzed using an integrated approach, an iterative process to determine themes and patterns present in the data.[45] The interviewers will develop the initial codebook together, and then independently dual code subsets of the transcripts to identify common themes and iteratively developed a final codebook, with periodic assessment of inter-rater reliability. We will follow the consolidated criteria for reporting qualitative research (COREQ) guidelines.[46]

Time to Event Analyses: As an exploratory analyses, we will also conduct time-to-event analyses for the primary outcome. Time-to-event will be defined as the interval from t = 0 to the first occurrence of hospitalization or emergency room visit, whichever occurs first. Participants without an event will be right-censored at the end of follow-up. Exploratory time-to-event analyses will be conducted using Cox proportional hazards regression, estimating hazard ratios with 95% confidence intervals. The proportional hazards assumption will be assessed using standard diagnostic methods (e.g., Schoenfeld residuals). Given the exploratory nature of this endpoint and the limited follow-up duration, results will be interpreted descriptively and in conjunction with the primary binary outcome analysis rather than as confirmatory evidence.

Sample size calculation: The primary outcome is the proportion of patients with a hospitalization or ER visit (all-cause) within 90 days. Preliminary data at the study sites suggests that at baseline, among a sample of eligible HF patients, 29% had a hospitalization rate or ER visit within a 90-day period; and based on prior data, we hypothesize a reduction to 13%.[8] We will therefore anticipate a sample size of 204 patients to detect a statistically significant difference with 80% power and an alpha error rate of 0.05, assuming no loss to follow up given the short duration of the study and very low loss to follow up rates in other trials at the study sites.[47]

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Pre-specified sub-group analyses: We will perform subgroup analysis only for the primary outcome based on the following pre-specified subgroups: EF (LVEF <50%, and >50%), gender, age (<65 and >=65 years), and food insecurity status (those with low or very low food security vs. food secure). For each subgroup, we will estimate group-specific absolute risks and relative risks using models consistent with the primary analysis. Effect modification will be assessed by including a treatment-by-subgroup interaction term in the regression model, with inference based on the interaction p-value. We will also analyze for significant changes in weight in those with obesity (BMI ≥ 30mg/kg2), HbA1c among those with diabetes (HbA1c ≥ 7), and cholesterol (total, triglycerides and LDL) among those with baseline dyslipidemia.

The trial is powered exclusively for the primary endpoint, and as such, all secondary, time-to-event, and subgroup analyses are designated exploratory, and will be considered as hypothesis-generating rather than confirmatory.

Discussion

As the first randomized trial of an Indigenous food is medicine program to improve cardiovascular health, MUTTON-HF will reveal important insights. This study will evaluate the impact of addressing food insecurity among patients with HF and if reclaiming traditional foods can advance Indigenous CV health. Additionally, this trial will evaluate the impact of food is medicine programs to not only improve patient cardiovascular health, but to also strengthen local food systems and advance Indigenous food sovereignty.

Cardiovascular inequities among the Indigenous populations are the direct result of the historic and ongoing impacts of settler colonialism. Precontact, Indigenous peoples in the US had a healthy lifestyle with consumption of traditional foods with demonstrated health benefits.[48–50] However, historic and ongoing genocide, structural violence, land theft, and exclusionary governmental policies, continue to perpetuate poverty and adverse social determinants of health in many tribal communities, which contribute to a disproportionate burden of cardiovascular disease.[1–4][51] Nutrition insecurity, due to restricted access to healthy food choices, contributes to poor cardiovascular health and outcomes.[4][52][53] Navajo Nation is a rural region of 27,000 square miles with ~175,000 residents. Navajo Nation is a food desert with high rates of household food insecurity.[4][54–56] Additionally, 1/3 of patients living on the reservation lack access to electricity and running water.[57] This has led to a reliance on ultra-processed foods and loss of food and plant diversity, which are harmful for health.[48][52][58–59]

Dietary factors have been shown to play an important role in HF, in particular.[7–9] Additionally, there is increasing evidence of the benefit of reclaiming traditional pre-contact foods for cardiometabolic health.[10–21] This study will be the first to evaluate if a CMTM program incorporating traditional foods can improve clinical and patient-related outcomes among an Indigenous population. We hypothesize that addressing food insecurity and decreasing reliance on ultra-processed foods through the provision of traditional healthy and sodium-restricted meals will improve disease-specific outcomes among patients with HF.

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Centering communities and protective assets of Indigenous communities is critical to advancing health in tribal communities. Beyond the benefit of sodium-restricted meals, we also hypothesize that MUTTON-HF will improve holistic health and well-being through strengthening cultural connectedness and addressing the interconnected determinants of Indigenous health.[1] This study will allow for evaluation of a community-designed strengths-based intervention grounded in community voice and storytelling – both essential to traditional knowledge systems.

Our pilot study demonstrated the feasibility and acceptability of an CMTM program, despite the rurality and vast distances between patients (spanning >100 miles apart).[24] Additionally, our pilot study showed promising results on the potential benefit on cultural connectedness, food security, and clinical outcomes, including components of the KCCQ scale, and significant weight loss among those with obesity. The proposed RCT will assess the clinical effectiveness of CMTMs among patients living with HF, including hospitalization rates, clinical outcomes, and quality of life.

This trial offers several unique aspects. This study will be, to our knowledge, the first study of locally supplied CMTMs for HF in both a rural and tribal setting. The majority of food is medicine trials evaluating MTM or MTGs have taken place in an urban setting, or where home delivery by large food corporations is utilized. [8][60–62] Our community-designed intervention is uniquely tailored to fit the local context by allowing for 1) sourcing of produce and meat from local farmers and ranchers 2) curated meal plan incorporating traditional, culturally-relevant recipes and foods, and 3) a delivery and implementation infrastructure to allow for delivery to patients living rurally and remotely, despite spanning vast differences, including those that postal service cannot reach and those that lack household infrastructure such as electricity. In addition to addressing a persistent evidence gap of the impact of Food is Medicine interventions on clinical and patient related outcomes, we believe that this trial will lend important implementation insights which will allow for more widespread adoption and scalability to other similar settings. Additionally, despite an increasing focus on advancing equity in HF care and outcomes, the cardiovascular health of the American Indian and Alaska Native population remains underprioritized, with few interventions directly designed and tested to improve Indigenous health.[46][63]. This intervention, and the subsequent outcomes tested by this study, will uniquely evaluate important aspects of Indigenous holistic health and well-being such as cultural connectedness, relationships with ancestral and other lands, and spiritual / emotional nourishment through the consumption of sacred traditional foods.[1] The effectiveness data, as well as data from the qualitative interviews, we anticipate will inform expansion and sustainable adoption of similar programs throughout the IHS both locally nationally.

This study will be limited by only including clinical sites in the Eastern Navajo Nation. However, the two sites include a larger IHS hospital and a satellite clinic which collectively serve communities located on and adjacent to the rural Navajo reservation and thus are likely representative of many IHS facilities. Additionally, inherent in the design as a pragmatic clinical trial to gain “real-world” experience, we included HF patients with an emergency room visit or hospitalization in the last 12 months. These inclusion criteria are broader than enrollment criteria in other HF trials including those that have evaluated MTMs.[8–

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10][59–62] It is possible that MTM may provide the most benefit in the post-discharge period, or that certain groups, such as those without other significant comorbidities (i.e. ESRD) might benefit the most from MTM. However, our inclusion criteria may offer greater generalizability and increase study efficiency by shortening the enrollment period. This CMTM intervention incorporating traditional foods will be compared to usual care. Given the lack of another intervention arm of non-culturally tailored MTMs, we will be unable to fully elucidate if the effectiveness of the MUTTON intervention is due to provision of traditional foods and recipes vs. solely due to provision of healthy food alone. However, this intervention was designed with community input to maximize acceptability, uptake, and engagement.[24]

In conclusion, we have undertaken a pragmatic randomized controlled trial that evaluates an important community-designed strategy to improve HF outcomes. MUTTON-HF will examine whether CMTMs incorporating locally sourced traditional Diné meals will improve clinical and patient-reported outcomes in HF in rural Navajo Nation. Lessons learned will provide valuable insight into the implementation and effectiveness of this Food is Medicine program for other tribal, as well as other rural communities, where food access is severely limited. If successful, this approach could be tailored and adopted in other tribal and rural settings to improve cardiovascular health.

Acknowledgements:

We acknowledge the patients with heart failure, the primary care physicians, the executive leadership committee of the Indian Health Service at the Gallup Indian Medical Center and the Tohatchi Health Center, the Eastern Navajo Tribal Council for their support. We acknowledge the Indian Health Service Office of Quality for their support and mentorship as part of our IHS Innovations Award, especially Stacy Hammer for her leadership and mentorship for Food is Medicine programs throughout the Indian Health Service. We thank the Gallup Community Food Pantry and staff, especially Lana Yazzie, Hilda Kendell, and Alice Perez for their support of this work. We also express our deep appreciation and gratitude to our farming and rancher partners. There was no financial compensation for these contributions.

Sources of Funding:

The MUTTON-HF trial was funded by the American Heart Association Healthcare by Food Initiative and the NHLBI, as well as through an Indian Health Service Innovations Award. The American Heart Association provided mentorship on the design and conduct of the study, but had no role in the collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Non-standard Abbreviations and Acronyms

DASH	Dietary Approaches to Stop Hypertension
DSQ	Dietary Screener Questionnaire
CCS	Cultural Connectedness Scale
CHR	Community Health Representative
CMTM	Culturally and Medically Tailored Meals
IHS	Indian Health Service
KCCQ	Kansas City Cardiomyopathy Questionnaire

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MTM	Medically Tailored Meals
MUTTON-HF	Medically Utilized Tailored Traditional food to Optimize Nutrition in Heart Failure
USDA FSSM	USDA Adult Food Security: Six Item short Form survey

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What is new?

- Settler colonialism has produced adverse structural drivers – particularly nutrition insecurity – that contribute to poor cardiovascular health among Indigenous populations.
- Reclaiming traditional pre-contact foods may be one mechanism to advance Indigenous health outcomes.
- Medically tailored meals have shown promise in improving heart failure outcomes and disease specific quality of life.
- In this context, utilizing community based participatory methods, we designed an Indigenous culturally and medically tailored meal (CMTM) program—MUTTON-HF (Medically Utilized Tailored Traditional food to Optimize Nutrition in Heart Failure) to improve health outcomes among patients with heart failure in rural Navajo Nation.

What are the clinical implications?

- The MUTTON-HF study will randomize patient with HF in rural Navajo Nation to 8 weeks of an Indigenous CMTM program versus usual care.
- MUTTON-HF will be the first study to evaluate the effectiveness of an Indigenous CMTM program on clinical and patient-reported outcomes.
- This trial will evaluate if an Indigenous CMTM program lowers the proportion of patients with hospitalization or emergency room visit (primary outcome).
- Secondary analyses will evaluate the impact of this community-designed CMTM program on disease-specific quality of life, HF symptoms, physical and social limitation, food insecurity, financial stress, Indigenous nourishment, and clinical biomarkers.

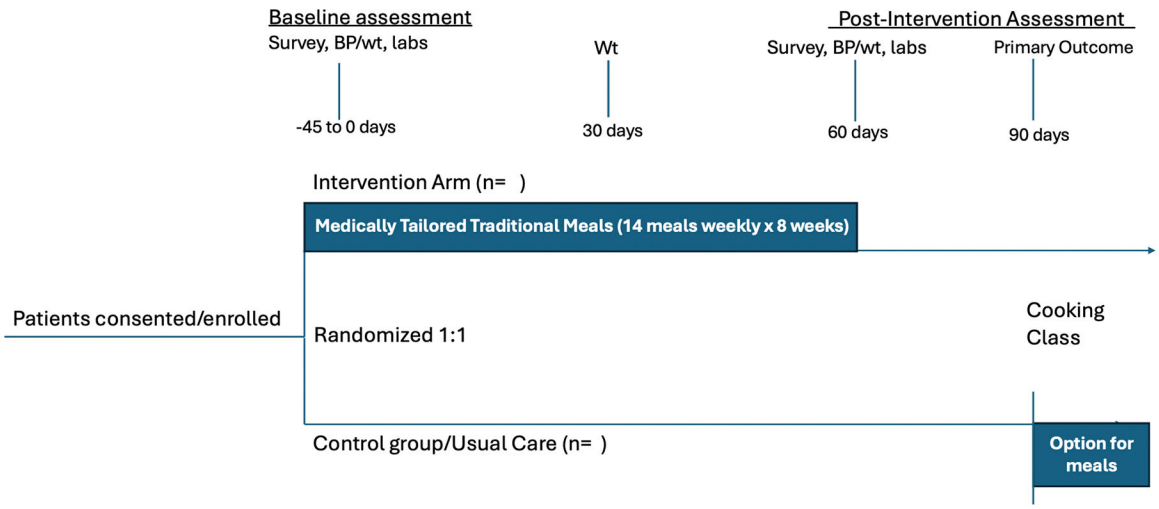


Figure 1. Study and Flow of Participants in Mutton-HF

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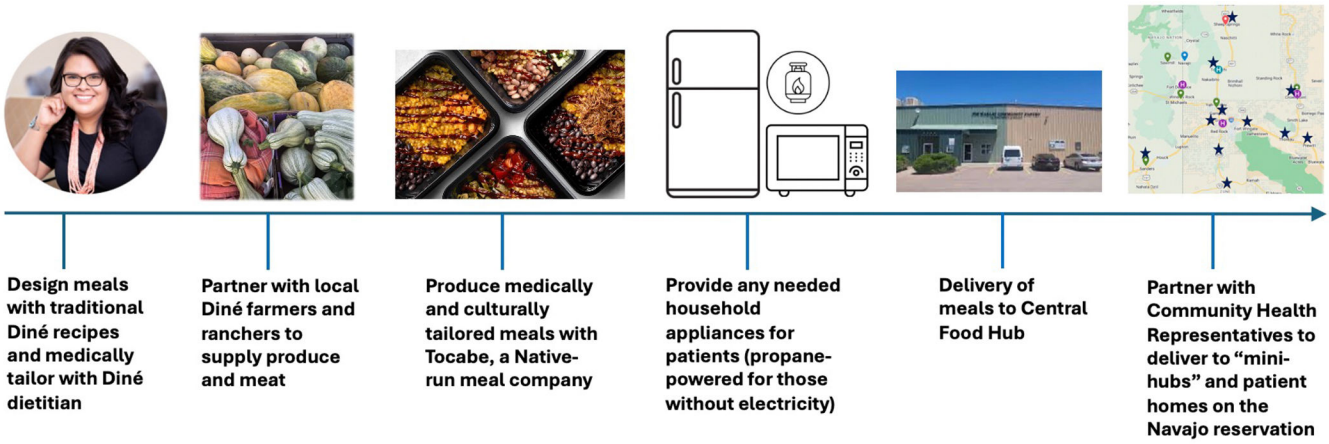


Figure 2. Overview of the MUTTON-HF Intervention

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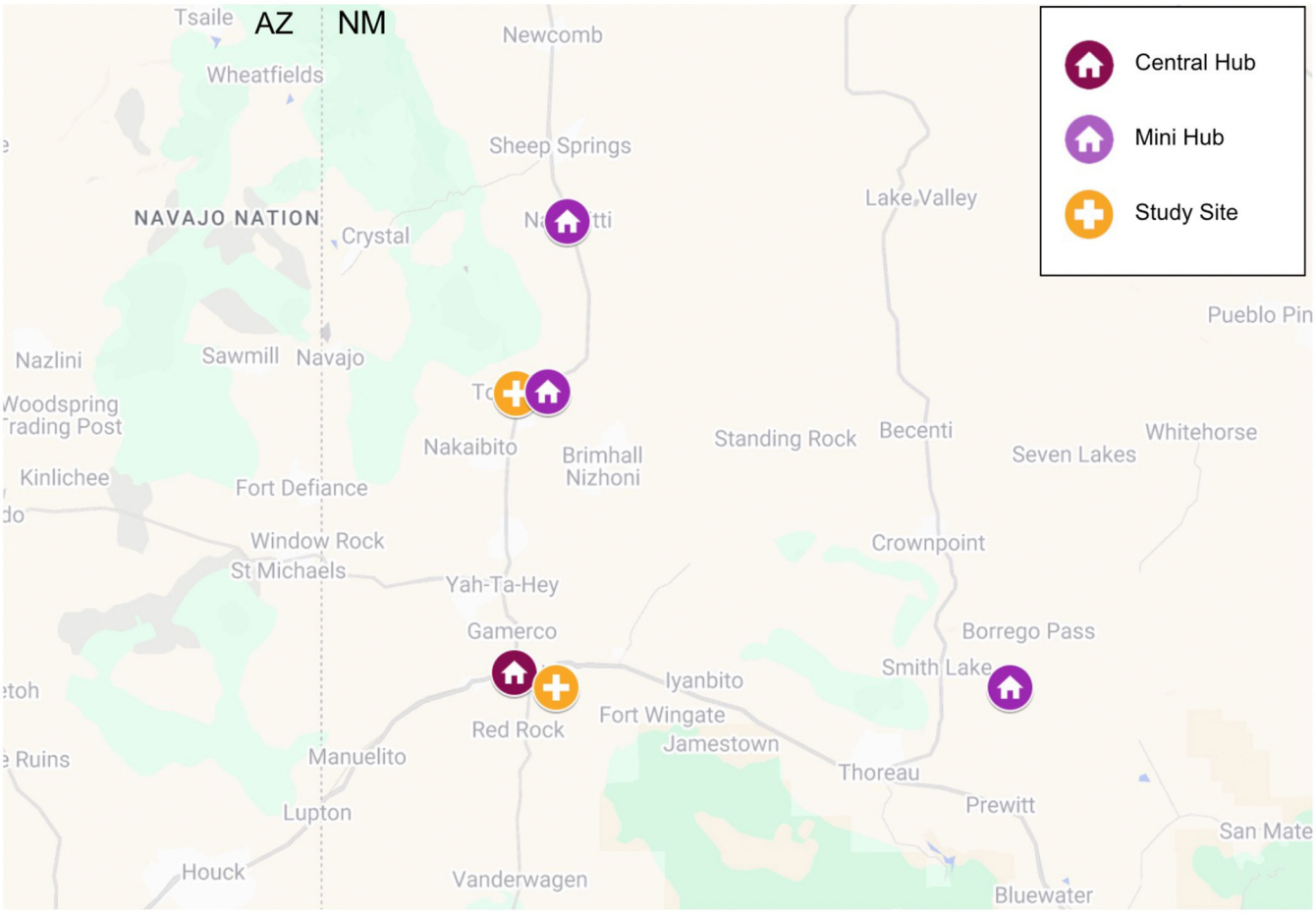


Figure 3. Map of Study Sites and Food Program Sites
AZ-Arizona; NM-New Mexico

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Table 1.

Primary and Secondary Outcomes of MUTTON-HF

Primary Outcome	Definition and Data Source	Data Source	Time Point
ER visit or Hospitalization	Clinical encounter in emergency room or hospitalization for any cause	Electronic Health Record	90 days
Secondary Outcomes			
ER visit or Hospitalization for Heart Failure	Clinical encounter in emergency room for heart failure or volume overload or Hospital Admission with Heart Failure as Primary Diagnosis	Electronic Health Record	90 days
ER visit for Heart Failure	Clinical encounter in emergency room for heart failure or volume overload as primary diagnosis	Electronic Health Record	90 days
Hospitalization for Heart Failure	Hospital Admission with Heart Failure as Primary Diagnosis	Electronic Health Record	90 days
Number of Hospitalizations and ER visits	Number of total clinical encounters in emergency room or hospitalizations for any cause	Electronic Health Record	90 days
KCCQ Total Score and Component Scores	Kansas City Cardiomyopathy Summary Score and its components including physical limitation score, symptom frequency score, quality of life score and social limitation score [33]	Patient survey *	Baseline, 60 days
Diet Quality *	Diet quality as assessed by the DSQ-10 [34]	Patient survey *	Baseline, 60-days
Food Security *	USDA Six Item Short Form survey [35]	Patient survey *	Baseline, 60 days
Indigenous Nourishment Scale *	Indigenous nourishment as measured by the Indigenous Nourishment Scale, Measure A [37]	Patient survey *	Baseline, 60 days
Financial Strain *	Financial Stress Scale [39]	Patient survey *	Baseline, 60 days
Cholesterol levels	Serum lipid panel to evaluate LDL cholesterol, triglycerides	Electronic Health Record	Baseline, 60 days
Glycemic Control	Serum HbA1c	Electronic Health Record	Baseline, 60 days
Weight	Weight	Home scale provided to patients	Baseline, 30 days, 60 days, 90 days
Blood pressure	Systolic and Diastolic Blood pressure	Home blood pressure cuff (Omron 5 series) provided to patients	Baseline, 60 days
Body Mass Index	Calculated Body Mass Index (kg/m2)	Patient height from electronic health record and weight from home scale provided to patients	Baseline, 30 days, 60 days, 90 days
Exploratory Outcomes			
Patient reported Health Status	Patient reported health status	Patient survey *	Baseline, 60 days
Exercise amount	Number of minutes of moderate-intensity or more of exercise weekly, as determined by self-report	Patient survey *	Baseline, 60 days
Safety			
Adverse Events	Death, acute kidney injury/worsening renal function, hyperkalemia, hypokalemia, hypoglycemia, hyperglycemia, hyponatremia, symptomatic hypotension	Electronic Health Record	Throughout

ER-emergency room; KCCQ- Kansas City Cardiomyopathy Questionnaire; DSQ-Diet Screener Questionnaire; USDA- United States Department of Agriculture; LDL-low-density lipoprotein; HbA1c-hemoglobin A1c.

* All patient-reported measures will be collected by patient survey over the phone or in person per patient preference.